

Attribution-quality metrics: MaRC and IG, baseline vs ABTT

200 random test pairs per model (up from 20 curated); LaTa + PhilTa.

Generated 2026-04-21.

Method	Suff@25% $\uparrow$		Comp@25% $\uparrow$		Cmpct@0.8 $\downarrow$		ρ_{LOO} $\uparrow$	
	base	abtt	base	abtt	base	abtt	base	abtt
<i>LaTa</i>								
IG	0.585	0.468	0.065	0.023	0.318	0.324	-0.014	0.106
MaRC	0.037	-0.014	-0.009	-0.001	0.902	0.903	-0.106	0.017
<i>PhilTa</i>								
IG	0.779	0.708	0.114	0.080	0.136	0.166	0.042	0.120
MaRC	-0.342	0.069	-0.009	0.019	0.661	0.916	-0.036	-0.120

Within each (method, metric) cell, the better of {baseline, ABTT} is **bolded**. Group means over 200 random test pairs per model. Standard errors are in the accompanying summary CSV; they shrank by $\sim \sqrt{10} \approx 3.2\times$ vs the 20-curated version, so differences of ~ 0.05 are now real.

Per-cell ABTT-wins count (out of 4 metrics):

- LaTa $\times$ IG: **1/4** (rho only)
- LaTa $\times$ MaRC: **2/4** (comp + rho)
- PhilTa $\times$ IG: **1/4** (rho only)
- PhilTa $\times$ MaRC: **2/4** (suff + comp; rho actually regresses)

Reading guide. **Suff@25%** ($\uparrow$): similarity retained with only the top 25% attributed tokens. **Comp@25%** ($\uparrow$): similarity drop when those tokens are removed. **Cmpct@0.8** ($\downarrow$): token fraction to recover 80% of full-pair cosine. **ρ_{LOO}** ($\uparrow$): rank correlation between attribution magnitude and true leave-one-out $\Delta \cos$.

Bottom line at 200 pairs/model.

- The original “ $\geq 3/4$ metrics for both MaRC and IG on at least one T5 model” threshold is still not met — no cell clears $3/4$.
- ρ_{LOO} now improves on **3/4** T5 cells (LaTa $\times$ IG, LaTa $\times$ MaRC, PhilTa $\times$ IG); PhilTa $\times$ MaRC regresses from -0.036 to -0.120 (a $\sim 4\sigma$ drop). So the “ABTT universally lifts ρ_{LOO} ” claim from the 20-pair snapshot no longer holds.
- The surviving robust signal: **for IG specifically**, ABTT lifts ρ_{LOO} on both T5 models, and the per-cell wins are $\geq 1/4$ on every cell we looked at. That is a much narrower framing than “ABTT improves explanation.”
- Caveat unchanged: cross-variant deltas conflate two effects (ABTT changes both the attribution scores and the cosine being explained), so these are descriptive shifts rather than fidelity-loss measures.